Sports clothing and athleisure attire is a huge industry, worth approximately $193 billion in 2021 with a strong growth forecast over the next decade!
(
`Source: https://www.statista.com/statistics/254489/total-revenue-of-the-global-sport`
)

In this workbook, you will undertake the role of a product analyst for an online sports clothing company. The company is specifically interested in how it can improve revenue. You will dive into product data such as pricing, reviews, descriptions, and ratings, as well as revenue and website traffic, to produce recommendations for its marketing and sales teams.

The data:

You've been provided with four datasets to investigate:

`brands.csv`

Columns	Description
`product_id`	Unique product identifier
`brand`	Brand of the product

`finance.csv`

Columns	Description
`product_id`	Unique product identifier
`listing_price`	Original price of the product
`sale_price`	Discounted price of the product
`discount`	Discount off the listing price, as a decimal
`revenue`	Revenue generated by the product

`info.csv`

Columns	Description
`product_name`	Name of the product
`product_id`	Unique product identifier
`description`	Description of the product

`reviews.csv`

Columns	Description
`product_id`	Unique product identifier
`rating`	Average product rating
`reviews`	Number of reviews for the product

```python
# Importing libraries
import pandas as pd

# Loading the data
brands = pd.read_csv("brands.csv")
finance = pd.read_csv("finance.csv")
info = pd.read_csv("info.csv")
reviews = pd.read_csv("reviews.csv")

# Merge the data and drop null values
merged_df = info.merge(finance, on="product_id")
merged_df = merged_df.merge(reviews, on="product_id")
merged_df = merged_df.merge(brands, on="product_id")
merged_df.dropna(inplace=True)

# Add price labels based on listing_price quartiles
merged_df["price_label"] = pd.qcut(merged_df["listing_price"], q=4, labels=
["Budget", "Average", "Expensive", "Elite"])
# Alternatively, avoiding pd.qcut(), get the individual quantiles and use pd.cut()
# twenty_fifth = merged_df["listing_price"].quantile(0.25)
# median = merged_df["listing_price"].quantile(0.5)
# seventy_fifth = merged_df["listing_price"].quantile(0.75)
# maximum = merged_df["listing_price"].max()
# merged_df["price_label"] = pd.cut(merged_df["listing_price"], bins=[0,
twenty_fifth, median, seventy_fifth, maximum], labels=["Budget", "Average",
"Expensive", "Elite"], include_lowest=True)

# Group by brand and price_label to get volume and mean revenue
adidas_vs_nike = merged_df.groupby(["brand", "price_label"], as_index=False).agg(
    num_products=("price_label", "count"),
    mean_revenue=("revenue", "mean")
).round(2)

print(adidas_vs_nike)

# Store the length of each description
merged_df["description_length"] = merged_df["description"].str.len()

# Upper description length limits
lengthes = [0, 100, 200, 300, 400, 500, 600, 700]

# Description length labels
labels = ["100", "200", "300", "400", "500", "600", "700"]

# Cut into bins
merged_df["description_length"] = pd.cut(merged_df["description_length"],
bins=lengthes, labels=labels)
```

```python
# Group by the bins
description_lengths = merged_df.groupby("description_length", as_index=False).agg(
    mean_rating=("rating", "mean"),
    total_reviews=("reviews", "sum")
).round(2)

print(description_lengths)
```

```
    brand price_label  num_products  mean_revenue
0  Adidas      Budget           574       2015.68
1  Adidas     Average           655       3035.30
2  Adidas   Expensive           759       4621.56
3  Adidas       Elite           587       8302.78
4    Nike      Budget           357       1596.33
5    Nike     Average             8        675.59
6    Nike   Expensive            47        500.56
7    Nike       Elite           130       1367.45
  description_length  mean_rating  total_reviews
0                100         2.26           36.0
1                200         3.19        17719.0
2                300         3.28        76115.0
3                400         3.29        28994.0
4                500         3.35         4984.0
5                600         3.12          852.0
6                700         3.65          818.0
```